

Threat Intelligence IOC Enrichment Tool

Professional Blue Team Project Documentation

1. Project Overview

The Threat Intelligence IOC Enrichment Tool is a Python-based blue team utility designed to automate the enrichment and analysis of Indicators of Compromise (IOCs) using the VirusTotal API. The project simulates a realistic SOC workflow by enabling analysts to validate suspicious indicators, assign threat severity levels, and generate structured intelligence reports suitable for triage and incident response investigations.

2. Project Objectives

- Automate IOC enrichment using real-world threat intelligence sources
- Support common IOC formats used in SOC operations and investigations
- Generate structured analyst-style threat intelligence reports
- Improve investigation efficiency during incident response workflows
- Demonstrate practical Python scripting and API integration skills

3. Key Features

- IOC type detection for IP addresses, domains, URLs, and file hashes
- VirusTotal API v3 integration
- Bulk IOC ingestion from text files
- Threat scoring and severity classification
- JSON and CSV report generation
- SOC-style analyst recommendations
- Error handling for invalid indicators and API rate limits
- Structured output suitable for operational workflows

4. Technologies and Libraries

Technology	Purpose
Python 3	Core scripting language
VirusTotal API v3	Threat intelligence enrichment
Requests	REST API communication
Pandas	CSV handling and reporting
JSON / CSV	Structured report generation

5. Threat Classification Logic

The tool assigns threat levels based on the number of malicious and suspicious verdicts returned by VirusTotal security vendors.

Threat Level	Detection Logic	Recommended Action
HIGH	20+ malicious detections	Block immediately and investigate related activity
MEDIUM	5+ malicious or suspicious detections	Investigate IOC and review related logs
LOW	Small number of detections	Perform manual validation before action
CLEAN / UNKNOWN	No significant detections	Monitor or validate context if required

6. Example Workflow

- 1 Analyst provides a suspicious IOC
- 2 Tool automatically detects IOC type
- 3 VirusTotal API enrichment is performed
- 4 Threat score and severity level are calculated
- 5 Structured reports are generated in JSON and CSV format
- 6 Analyst recommendations are produced for triage or escalation

7. Command Execution Examples

Single IOC analysis:

```
python threat_intel_enricher.py -i 8.8.8.8
```

Bulk IOC analysis:

```
python threat_intel_enricher.py -f iocs.txt
```

8. Sample Detection Output

```
IOC: 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
IOC Type: file_hash
Malicious Detections: 66
Suspicious Detections: 0
Threat Level: HIGH
Recommendation: Block immediately and investigate related activity.
```

9. Project Repository Structure

```
ioc-threat-intel-enricher/
■■■ threat_intel_enricher.py
■■■ requirements.txt
■■■ iocs.txt
■■■ threat_intel_report.json
■■■ threat_intel_report.csv
■■■ README.md
■■■ screenshots/
```

10. SOC and Blue Team Use Cases

- Threat intelligence enrichment
- IOC triage during incident response
- Malware hash reputation analysis
- Threat hunting preparation
- SOC alert validation workflows
- Security operations reporting support

11. Future Improvements

- ANY.RUN sandbox integration
- MITRE ATT&CK; mapping support
- AbuseIPDB integration
- HTML and PDF report generation
- Email alerting functionality

- SIEM integration support

12. Conclusion

This project demonstrates practical blue team and threat intelligence capabilities using Python and real-world threat intelligence APIs. It reflects operational SOC workflows involving IOC enrichment, malware reputation analysis, threat scoring, reporting automation, and investigation support. The project was designed to simulate realistic analyst tasks commonly performed in modern security operations environments.